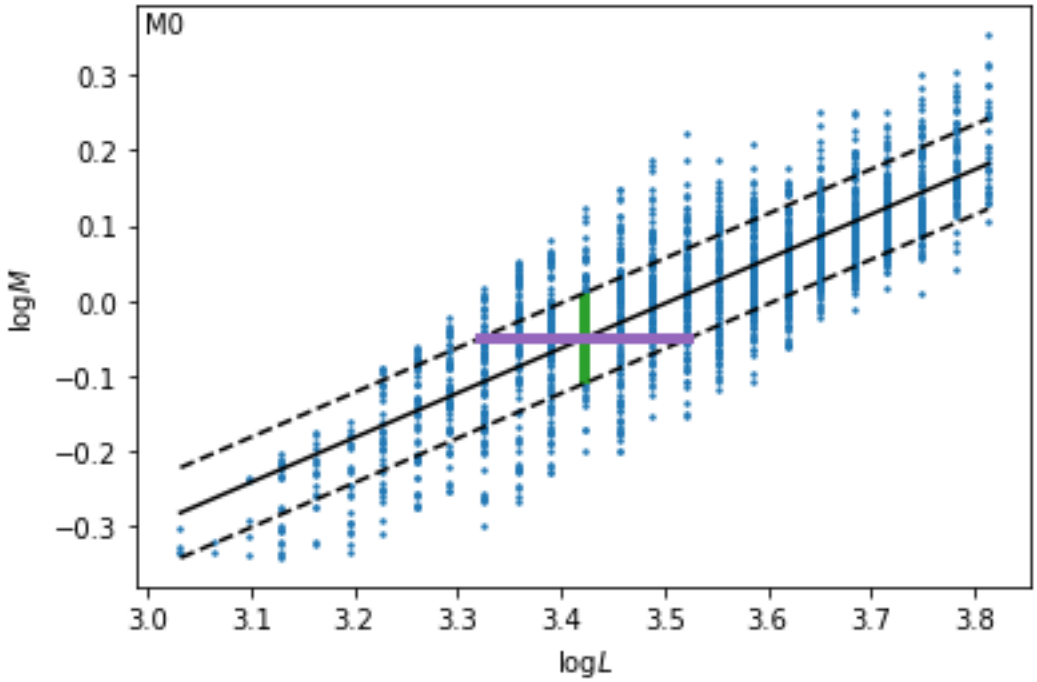
M0:

LM Strip:

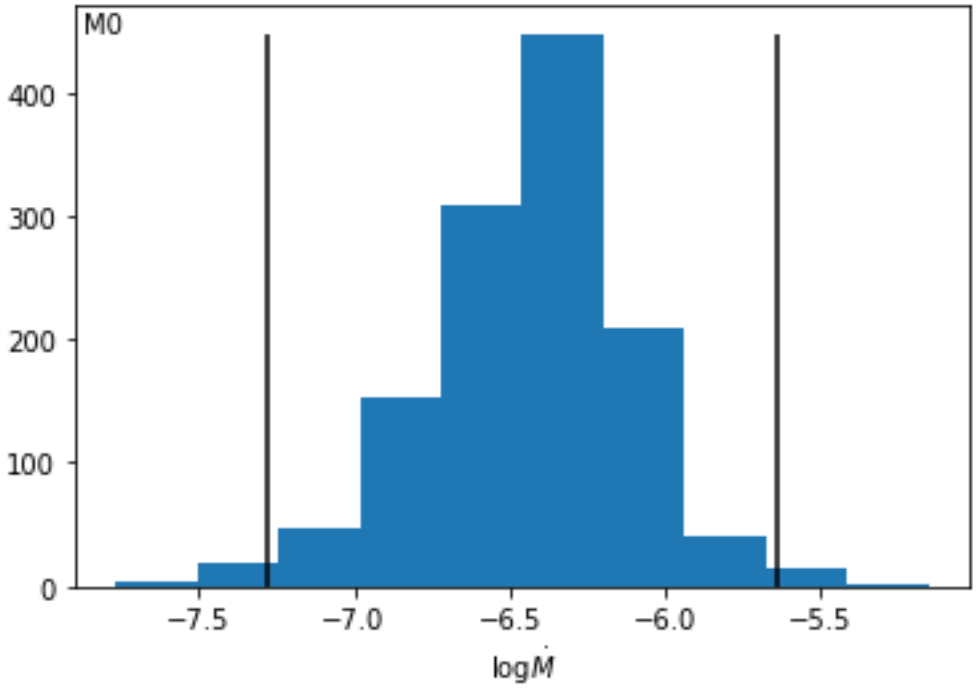
deathline_fit: (array([-2.08166269, 0.59359611]), 0.6939735315292701, Intercept 0.031530
x   0.008929
dtype: float64,   Intercept   x
Intercept   0.233686 -0.066104
x   -0.066104 0.018740)
Optimal fit found at 5694
n= 1.0693999999999373
Fit quality: Slope= 1.0000037850481107
Fit significance: R^2= 0.19180011520815876
Fit standard dev: sigma= 0.802789559750684
delta logL, delta logM, delta logMdot: [0.20198361631524042, 0.11989668960937344, 1.6423545861871354]
log A, B, C: [-34.975963392304195, 8.131127742677283, -13.698081169196328]
Stars in Bound: 1243

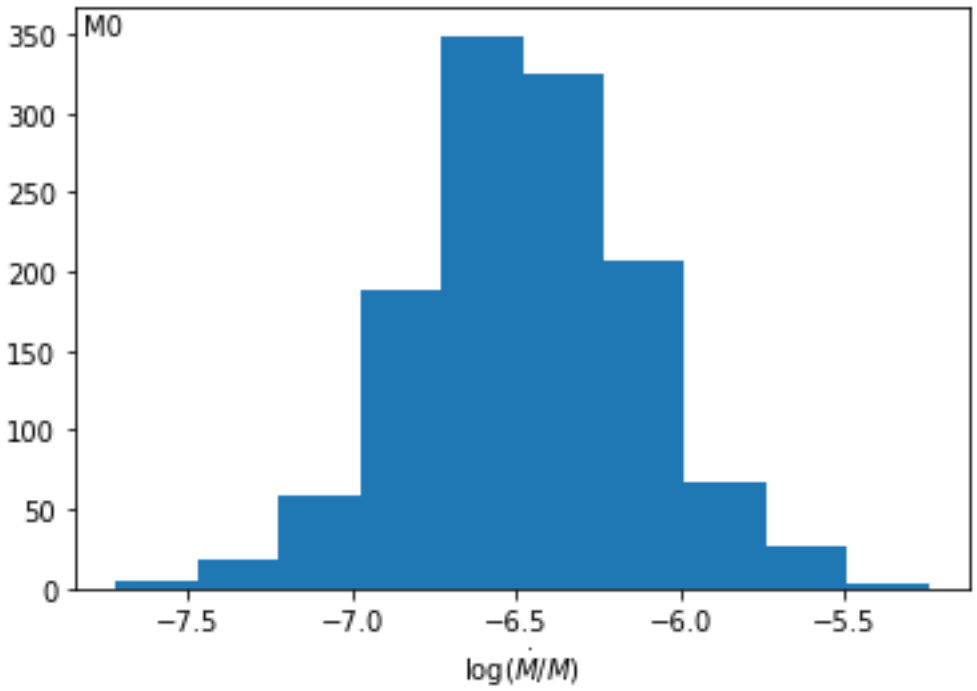
Lower bound fit:
n= 0.5
Fit quality: Slope= 2.12430541120037
Fit significance: R^2= 0.1918001152081602
Fit standard dev: sigma= 1.7611346721938628
delta logL, delta logM, delta logMdot: [0.09443782322575789, 0.05605792482204064, 1.6312182200691758]
log A, B, C: [-67.03506056677335, 17.272933284047376, -29.09879781043589]
Stars in Bound: 659

Upper bound fit:
n= 2.999900000001945
Fit quality: Slope= 0.4093764453350075
Fit significance: R^2= 0.1918001152081592
Fit standard dev: sigma= 0.4016934426344642
delta logL, delta logM, delta logMdot: [0.5666080517902725, 0.33633633734749735, 1.886056647693164]
log A, B, C: [-18.13441585975965, 3.3286795726497718, -5.607650551728879]
Stars in Bound: 1938




```
Full range: -5.155448273660727 -7.7648075037301885
vlines locations: -5.638032424081891 -7.280387010269027
```





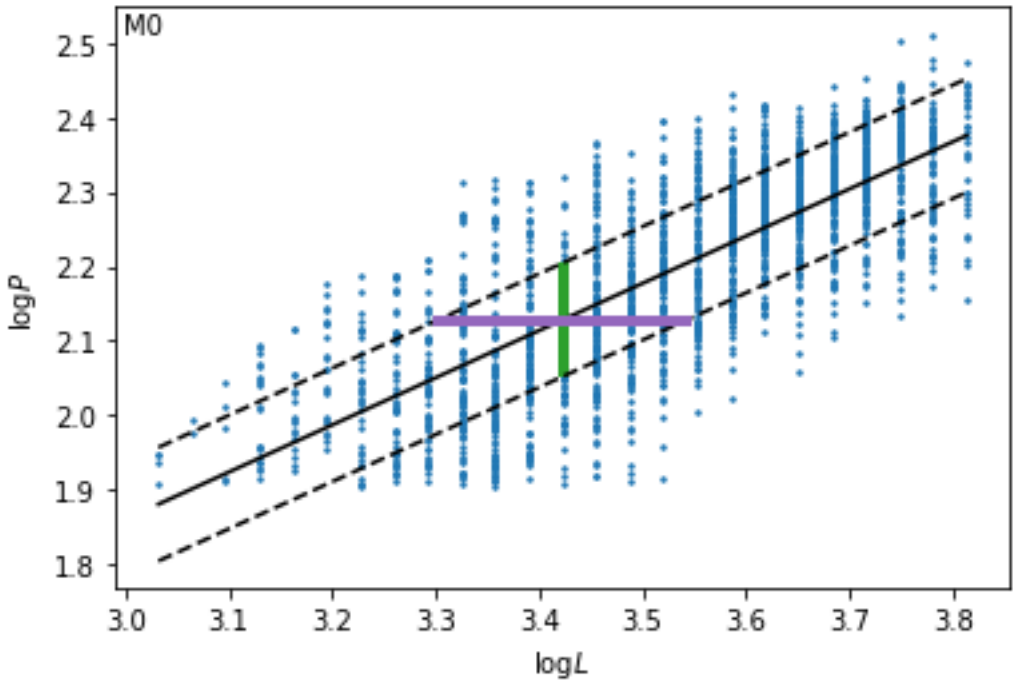
LP Strip:

deathline_fit: (array([-0.03909448, 0.63325243]), 0.6084726695644294, Intercept 0.040631
x   0.011506
dtype: float64,   Intercept   x
Intercept   0.233686 -0.066104
x   -0.066104 0.018740)
Optimal fit found at 5729
n= 1.072899999999937
Fit quality: Slope= 1.0000078231403786
Fit significance: R^2= 0.19180011520815932
Fit standard dev: sigma= 0.8027928014811921
delta logL, delta logM, delta logMdot: [0.24049875653099395, 0.15229642227930285, 1.618873490337612]
log A, B, C: [-6.045614284886297, -6.731317507369241, 10.62975391088762]
log A, B, C (LM Strip): [-34.98519371555522, 8.135775205015511, -13.701508306794192]
Stars in Bound: 1227

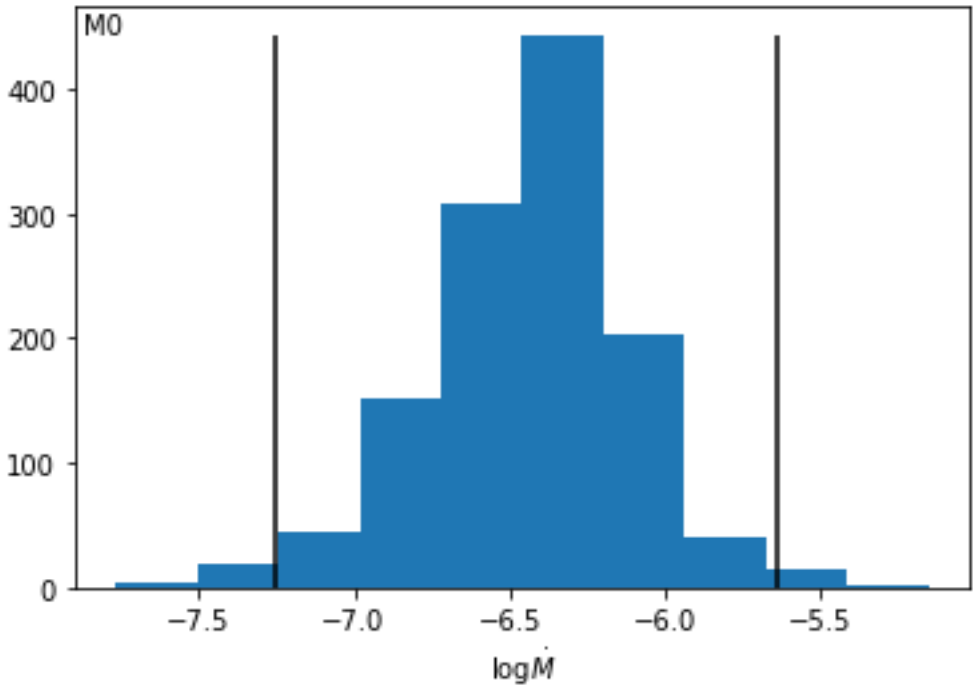
Lower bound fit:
n= 0.5
Fit quality: Slope= 2.1310555337740062
Fit significance: R^2= 0.1918001152081591
Fit standard dev: sigma= 1.7670414586499072
delta logL, delta logM, delta logMdot: [0.11207883145260888, 0.07097419250597081, 1.6077371242196525]
log A, B, C: [-5.575594500661207, -14.344699203073542, 22.652418681401684]
Stars in Bound: 647

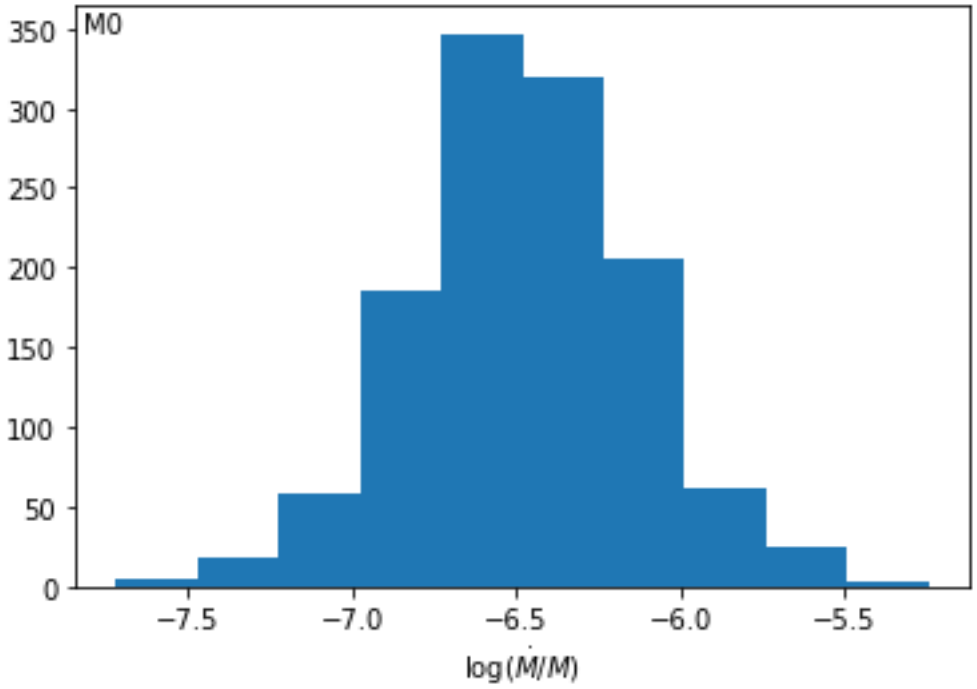
Upper bound fit:
n= 2.999900000001945
Fit quality: Slope= 0.4166752321604381

```
Fit significance: R^2= 0.1918001152081591
Fit standard dev: sigma= 0.4048860864073366
delta logL, delta logM, delta logMdot: [0.6724505729497987, 0.42583096019759914,
1.886056647693164]
log A, B, C: [-6.288024800345623, -2.8047513431652114, 4.429120528995763]
# Stars in Bound: 1936
```




```
Full range: -5.155448273660727 -7.7648075037301885
vlines locations: -5.638032424081891 -7.256905914419503
```





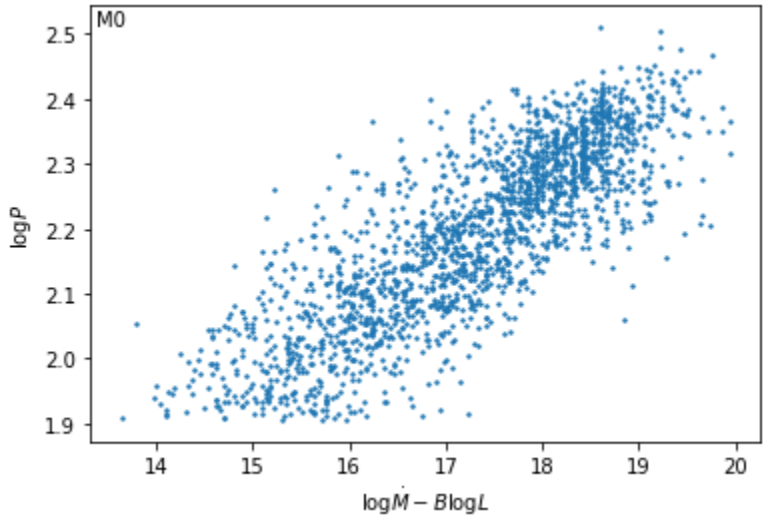
Comparison of Exponents:

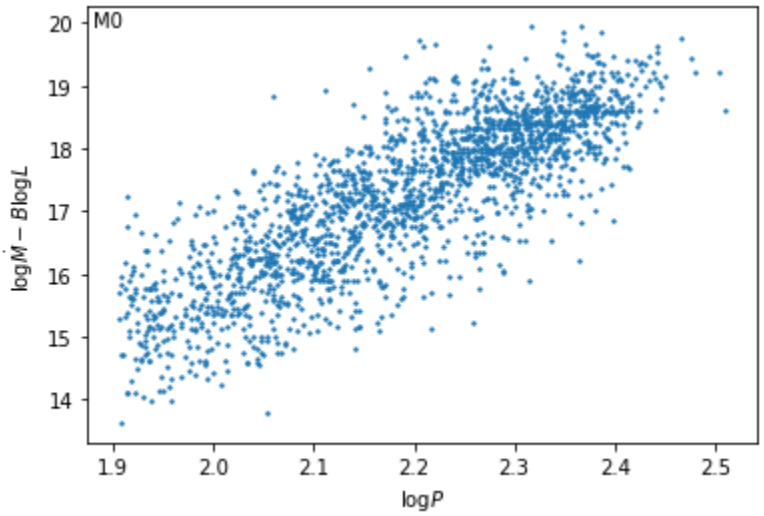
LM Strip:

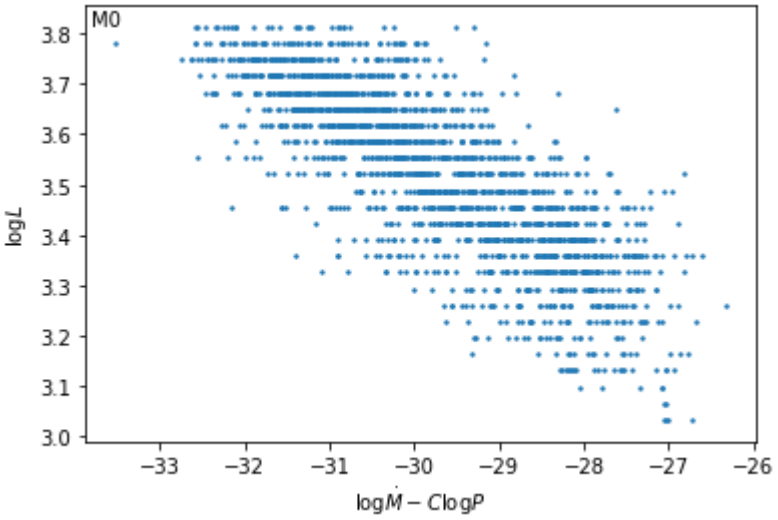
log A, B, C: [-34.975963392304195, 8.131127742677283, -13.698081169196328]

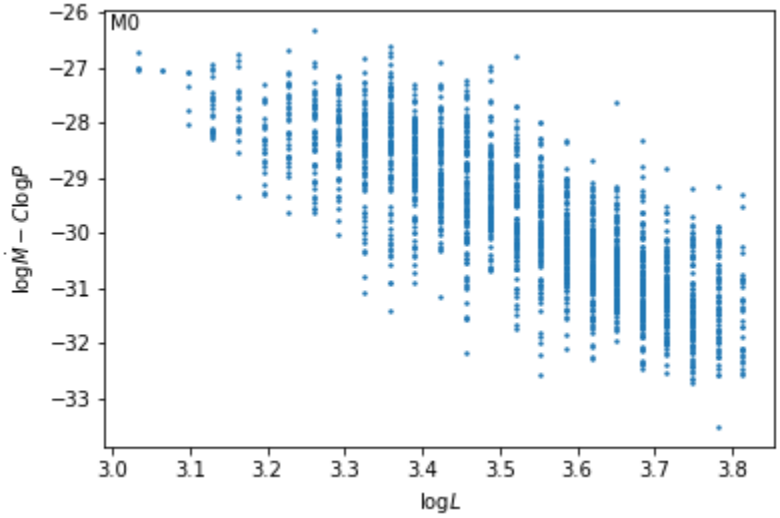
LP Strip:

log A, B, C: [-6.045614284886297, -6.731317507369241, 10.62975391088762]
log A, B, C (LM Strip): [-34.98519371555522, 8.135775205015511, -13.701508306794192]




M0
20
19
18
17
16
15
14
1.9
2.0
2.1
2.2
2.3
2.4
2.5
$\log\dot{M} - B\log L$
$\log P$


M0
3.8
3.7
3.6
3.5
3.4
3.3
3.2
3.1
3.0
−33
−32
−31
−30
−29
−28
−27
−26
$\log L$
$\log\dot{M} - C\log P$

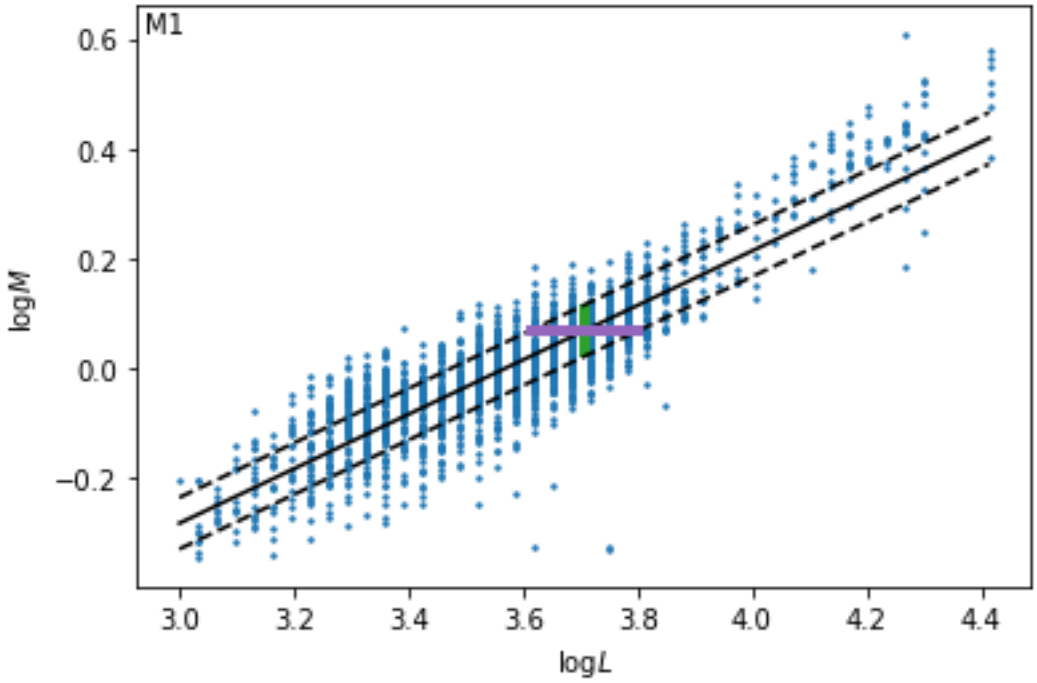

M0
−26
−27
−28
−29
−30
−31
−32
−33
3.0
3.1
3.2
3.3
3.4
3.5
3.6
3.7
3.8
$\log\dot{M} - C\log P$
$\log L$

M1:
LM Strip:

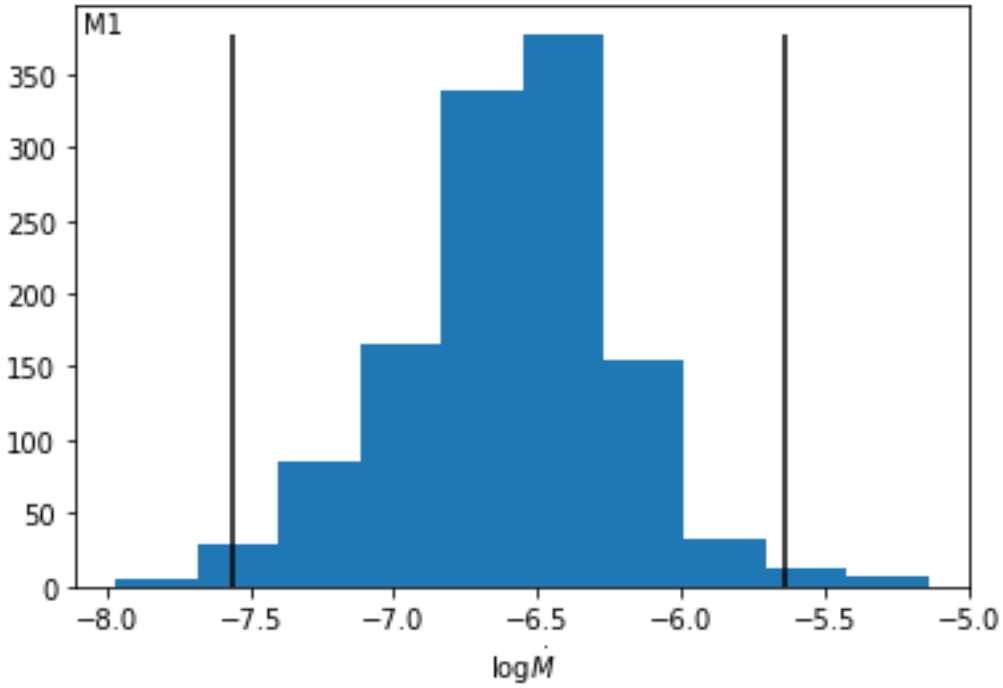
deathline_fit: (array([-1.77015472, 0.4958621]),
0.7647445924769394, Intercept   0.021242
x   0.005918
dtype: float64,   Intercept   x
Intercept   0.117549 -0.032683
x   -0.032683 0.009123)
Optimal fit found at 3451
n= 0.845099999999962
Fit quality: Slope= 0.9999967327250667
Fit significance: R^2= 0.12227636713260515
Fit standard dev: sigma= 1.1878406074267007
delta logL, delta logM, delta logMdot: [0.1891101235031245, 0.09377254366392473, 1.9199034860015933]
log A, B, C: [-42.82979371999287, 10.152304120143375, -20.474047210262466]
Stars in Bound: 1207

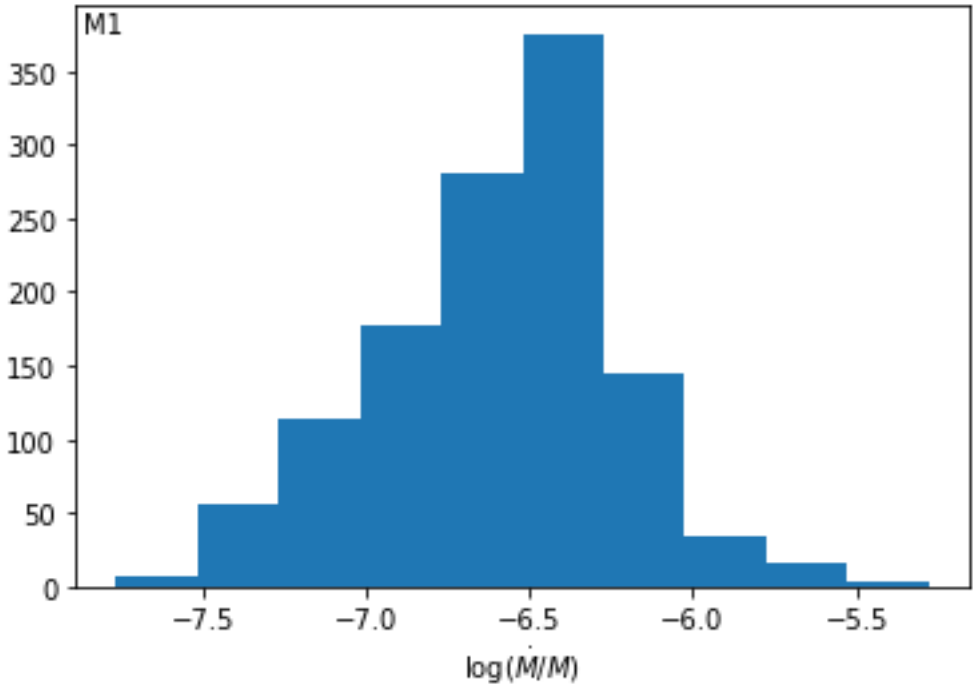
Lower bound fit:
n= 0.5
Fit quality: Slope= 1.5986696535219758
Fit significance: R^2= 0.12227636713260459
Fit standard dev: sigma= 1.9174306237799505
delta logL, delta logM, delta logMdot: [0.11188624038760686, 0.05548014652936273, 1.8159398112730436]
log A, B, C: [-64.52710704694141, 16.230233538834568, -32.7313449021257]
Stars in Bound: 781

Upper bound fit:
n= 2.999900000001945
Fit quality: Slope= 0.2956713860252729
Fit significance: R^2= 0.12227636713260315
Fit standard dev: sigma= 0.46995832024867795
delta logL, delta logM, delta logMdot: [0.6712950650780005, 0.3328697831470864, 2.0150637464358763]
log A, B, C: [-17.30338823114128, 3.001755638113827, -6.053609695012397]
Stars in Bound: 2150




```
Full range: -5.137719506700295 -7.967457791305338
vlines locations: -5.638032424081891 -7.557935910083485
```





LP Strip:

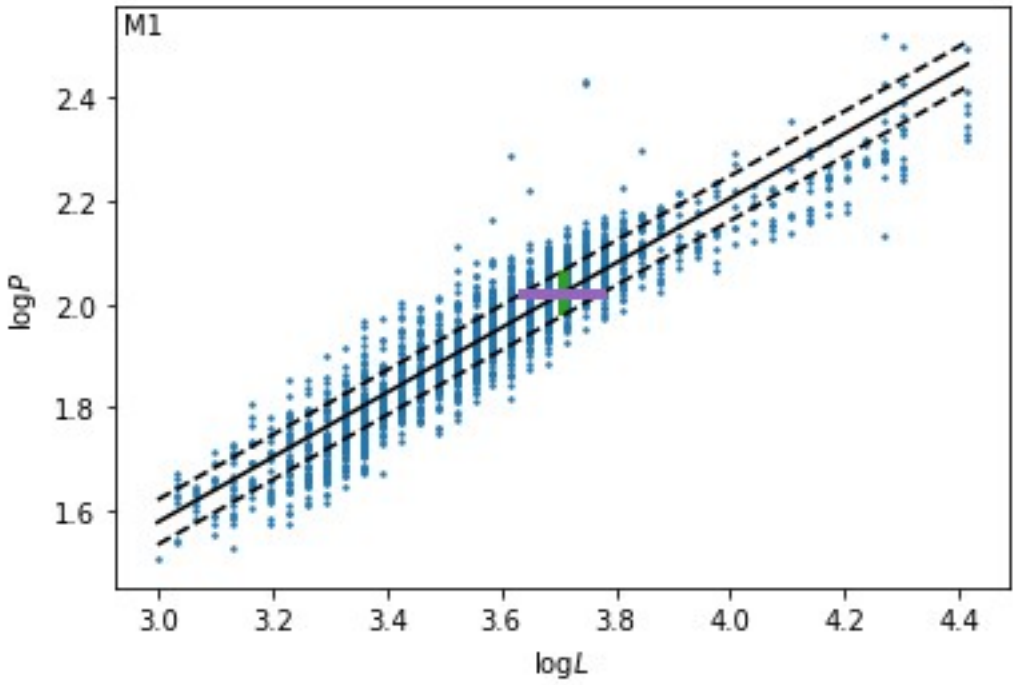
deathline_fit: (array([-0.29918172, 0.62587935]), 0.8587857374171389, Intercept   0.019602

```
x           0.005461
dtype: float64,           Intercept         x
Intercept   0.117549 -0.032683
x          -0.032683  0.009123)
Optimal fit found at  3932
n= 0.8931999999999567
Fit quality: Slope= 0.999992410574981
Fit significance: R^2= 0.12227636713260481
Fit standard dev: sigma= 1.1878354733884267
delta logL, delta logM, delta logMdot: [0.13826171036643942, 0.0865351490956483,
1.9199034860015933]
log A, B, C: [0.05020310565563323, -13.886009951079094, 22.18640062524769]
log A, B, C (LM Strip): [-42.822131870554244, 10.148517846251726, -
20.479600825147386]
# Stars in Bound: 1207

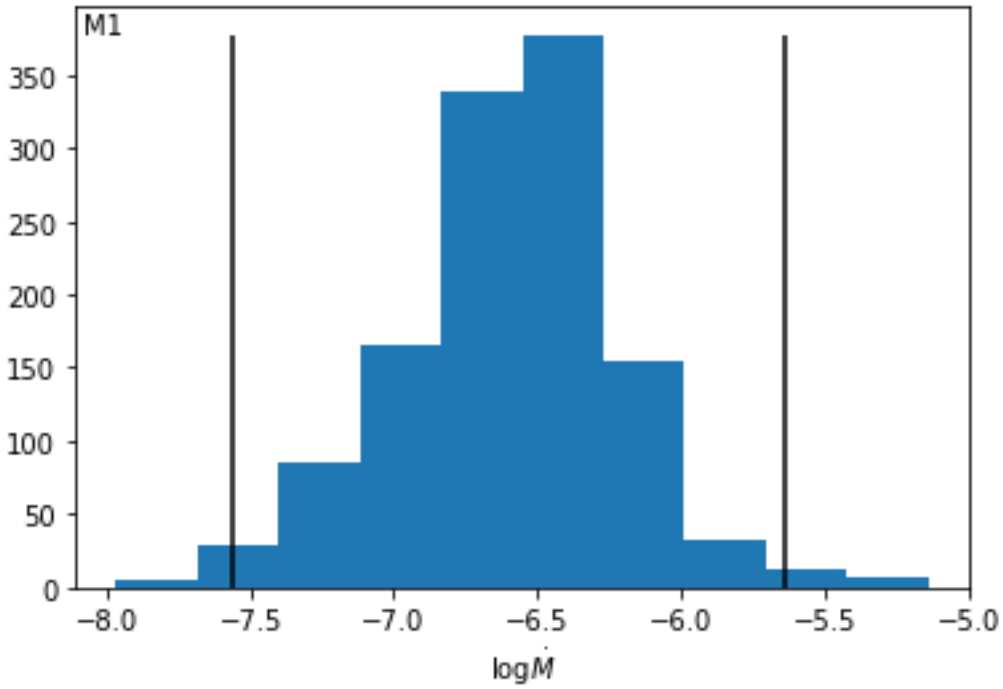

Lower bound fit:
n= 0.5
Fit quality: Slope= 1.7488940794957306
Fit significance: R^2= 0.12227636713260648
Fit standard dev: sigma= 2.1037801990542824
delta logL, delta logM, delta logMdot: [0.07739683741964054, 0.048441082118031886,
1.8796088911424205]
log A, B, C: [5.021274554207403, -24.285344903065035, 38.80195918337563]
# Stars in Bound: 735

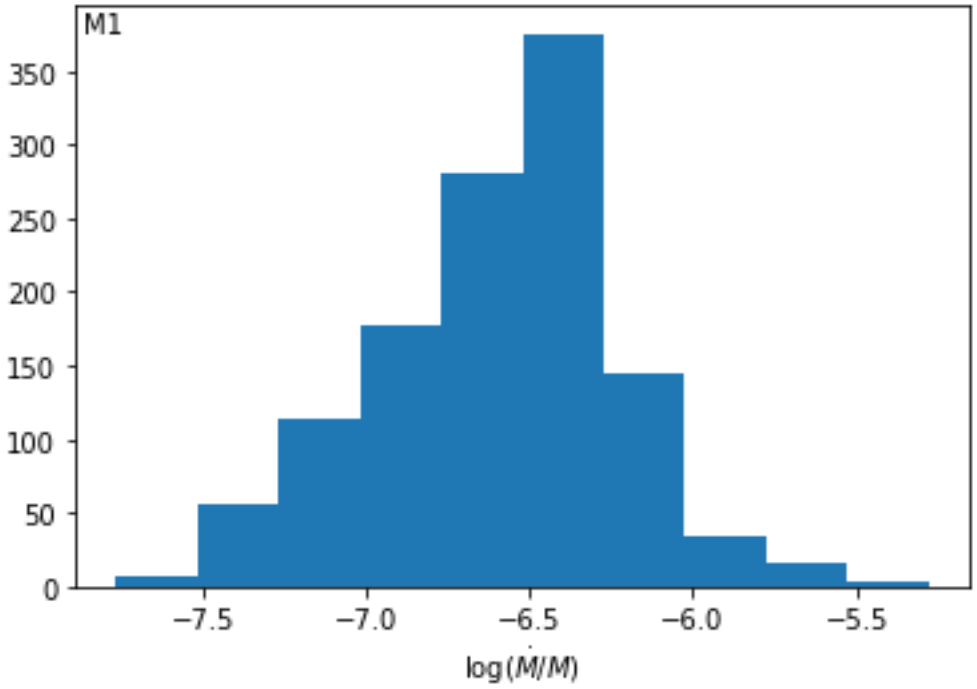

Upper bound fit:
n= 2.999900000001945
Fit quality: Slope= 0.31249856885754446
```

```
Fit significance: R^2= 0.12227636713260437
Fit standard dev: sigma= 0.48030906064683926
delta logL, delta logM, delta logMdot: [0.46436554515066275, 0.2906368044919585,
2.0150637464358763]
log A, B, C: [-4.513254487571495, -4.339391170337781, 6.933271063030939]
# Stars in Bound: 2141
```




```
Full range: -5.137719506700295 -7.967457791305338
vlines locations: -5.638032424081891 -7.557935910083485
```





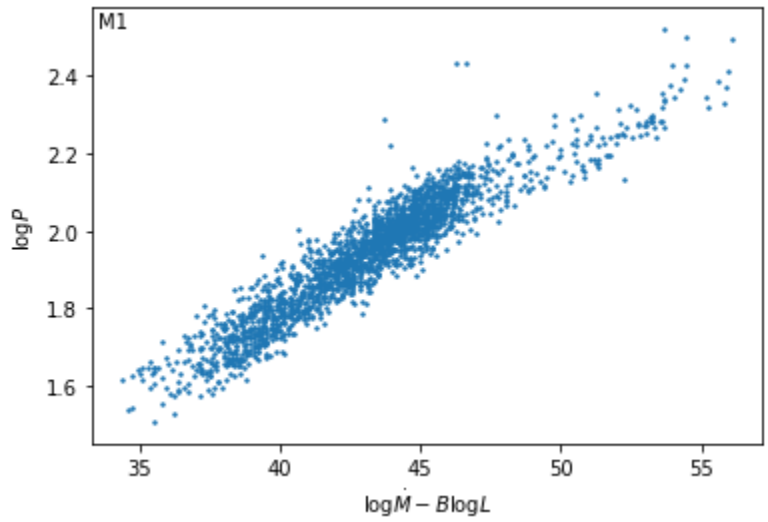
Comparison of Exponents:

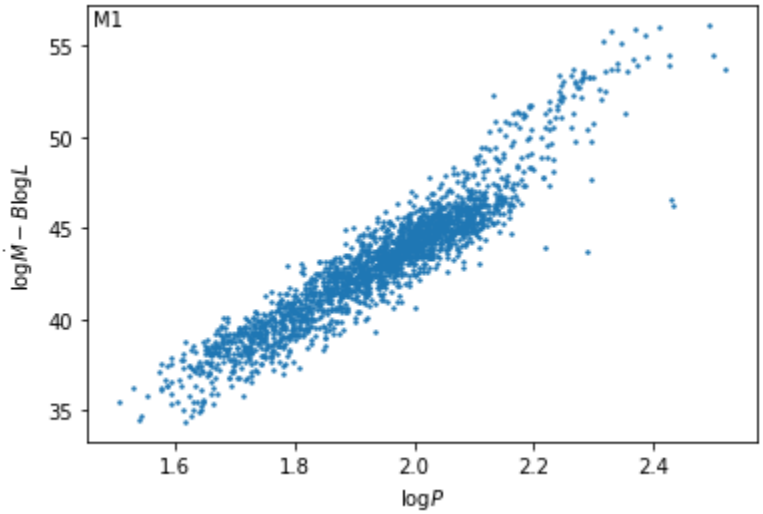
LM Strip:

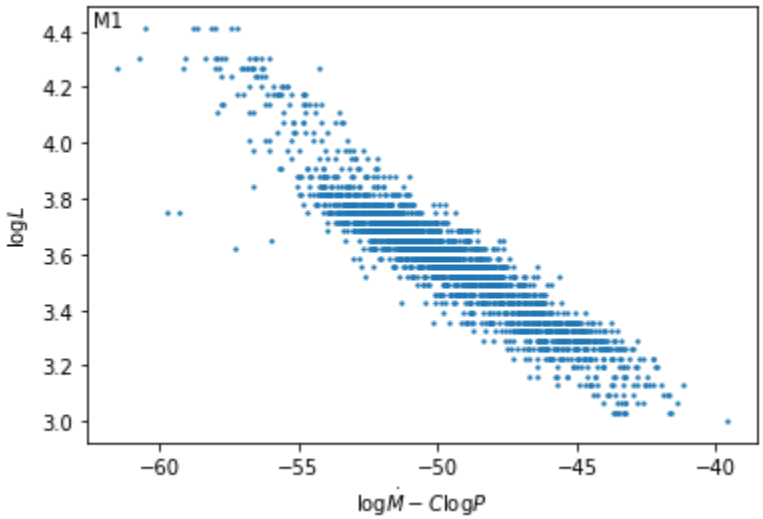
```
log A, B, C: [-42.82979371999287, 10.152304120143375, -20.474047210262466]
```

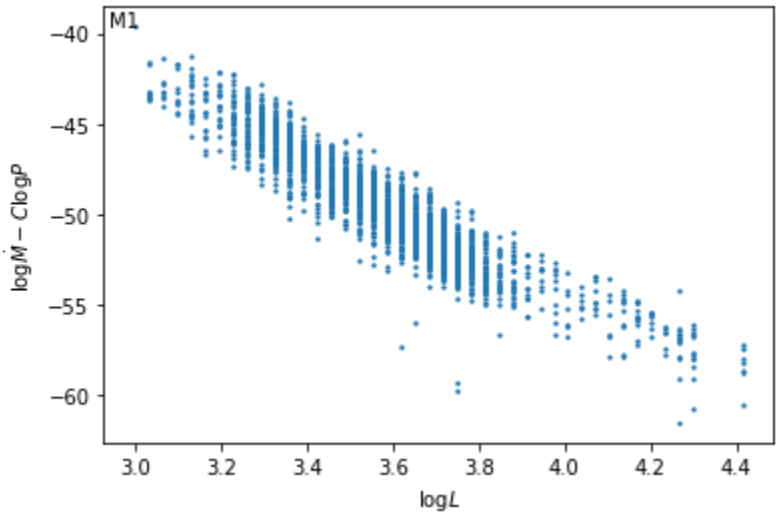
LP Strip:

```
log A, B, C: [0.05020310565563323, -13.886009951079094, 22.18640062524769]
log A, B, C (LM Strip): [-42.822131870554244, 10.148517846251726, -
20.479600825147386]
```




M1
55
50
45
40
35
1.6
1.8
2.0
2.2
2.4
$\log\dot{M} - B\log L$
$\log P$


M1
4.4
4.2
4.0
3.8
3.6
3.4
3.2
3.0
−60
−55
−50
−45
−40
$\log L$
$\log\dot{M} - C\log P$


M1
−40
−45
−50
−55
−60
3.0
3.2
3.4
3.6
3.8
4.0
4.2
4.4
$\log\dot{M} - C\log P$
$\log L$

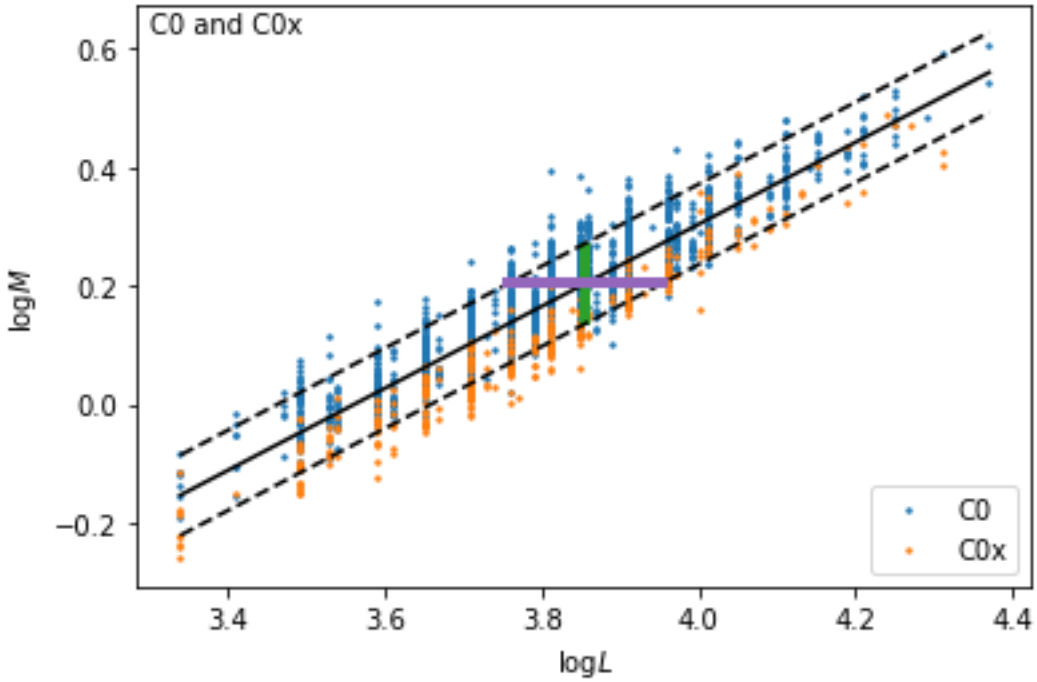
C0:

LM Strip:

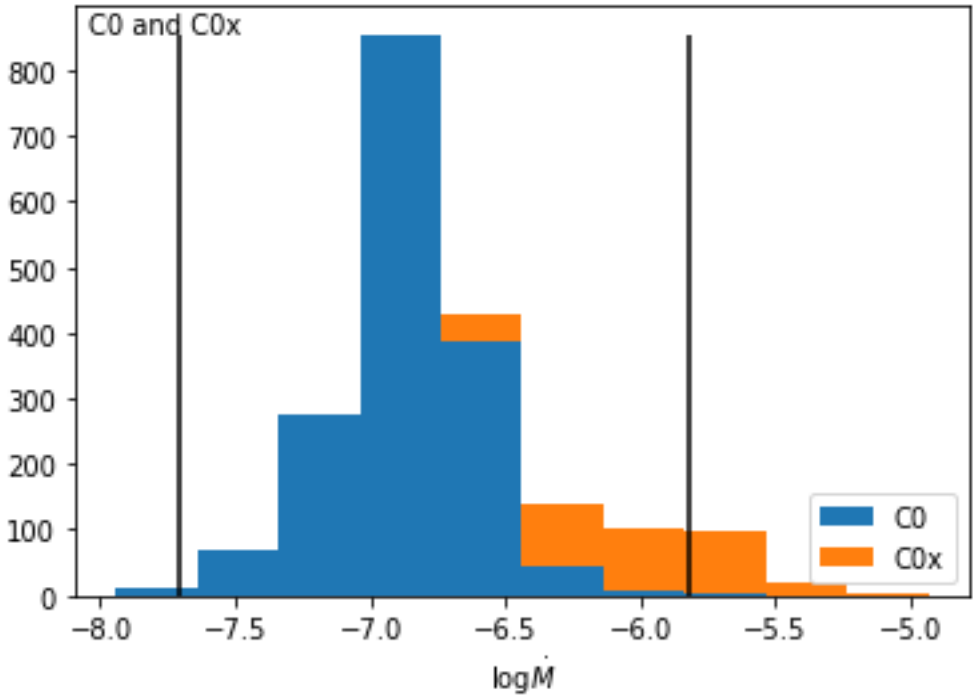
deathline_fit: (array([-2.46712362, 0.69265989]), 0.8425134640215446,
Intercept   0.025548
x   0.006710
dtype: float64,   Intercept   x
Intercept   0.279136 -0.073246
x   -0.073246 0.019255)
Optimal fit found at 12040
n= 1.7039999999998674
Fit quality: Slope= 1.000029174272829
Fit significance: R^2= 0.4073840902941994
Fit standard dev: sigma= 0.5175468273335767
delta logL, delta logM, delta logMdot: [0.19548072568275288,
0.13540165823092454, 1.8834944309285628]
log A, B, C: [-41.07893702614187, 9.635192545710618, -13.910423664947327]
Stars in Bound: 1668

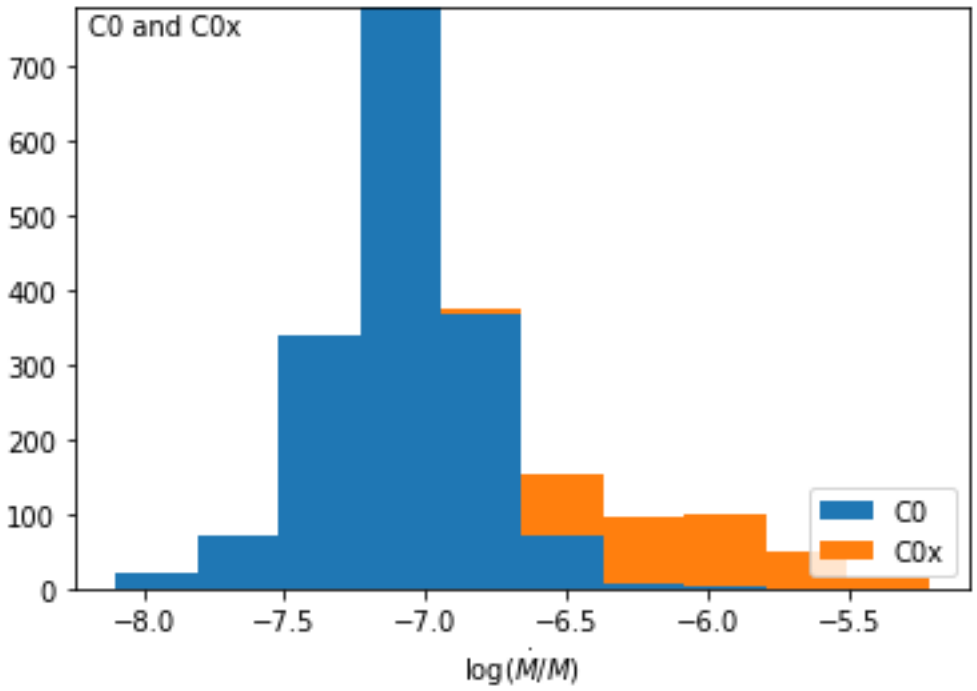
Lower bound fit:
n= 0.5
Fit quality: Slope= 2.5481476018874867
Fit significance: R^2= 0.407384090294197
Fit standard dev: sigma= 1.4766142733923031
delta logL, delta logM, delta logMdot: [0.057359367864662225,
0.03973053351846678, 1.40823996531185]
log A, B, C: [-94.20685284118935, 24.551176516354776, -35.444778627432704]
Stars in Bound: 633

Upper bound fit:
n= 2.999900000001945
Fit quality: Slope= 0.6106583959313419
Fit significance: R^2= 0.40738409029419953
Fit standard dev: sigma= 0.3574753134983903
delta logL, delta logM, delta logMdot: [0.3441447353146221, 0.238375255004252,
2.0248235837250323]
log A, B, C: [-27.716614374612327, 5.883639573547186, -8.494269187835435]
Stars in Bound: 1970




```
Full range: -4.933674074637962 -7.942447998945467
vlines locations: -5.819730722331125 -7.703225153259688
```





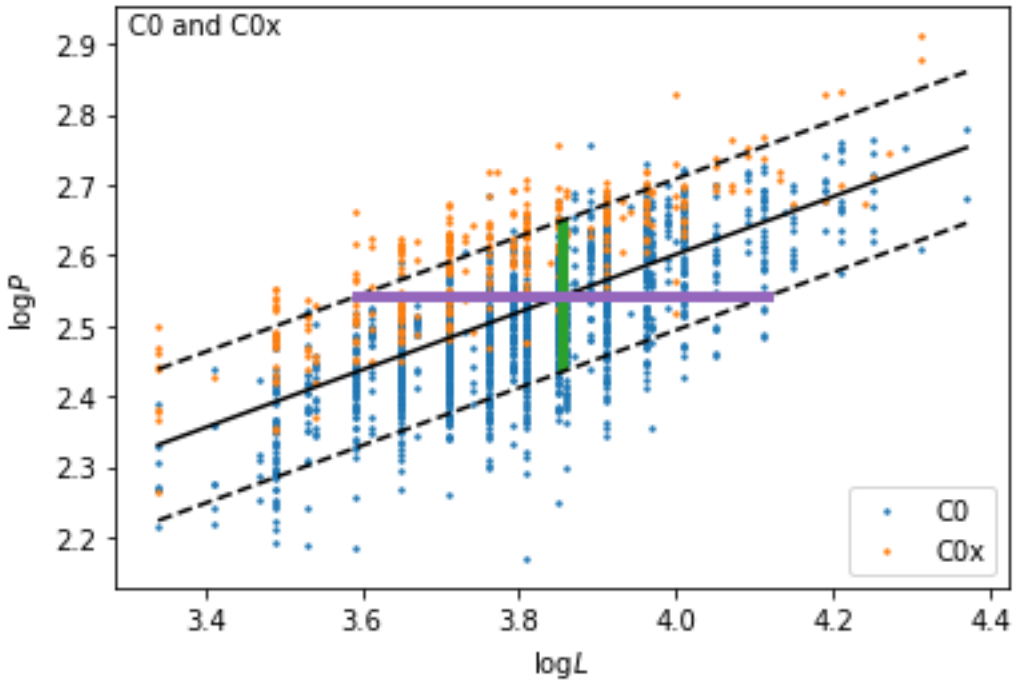
LP Strip:

```
deathline_fit: (array([0.96374584, 0.4094138 ]), 0.42666412702887635,
Intercept    0.040487
x           0.010634
dtype: float64,           Intercept         x
Intercept   0.279136 -0.073246
x          -0.073246  0.019255)
Optimal fit found at  10137
n= 1.5136999999998884
Fit quality: Slope= 0.9999882712893458
Fit significance: R^2= 0.40738409029419886
Fit standard dev: sigma= 0.5175256586149053
delta logL, delta logM, delta logMdot: [0.5241431436444071, 0.214591436810891,
1.8834944309285628]
log A, B, C: [-15.219113434240555, -3.5934733741482967, 8.777118318045444]
log A, B, C (LM Strip): [-41.06675262075109, 9.622154683267865, -
13.902542891223035]
# Stars in Bound: 1668

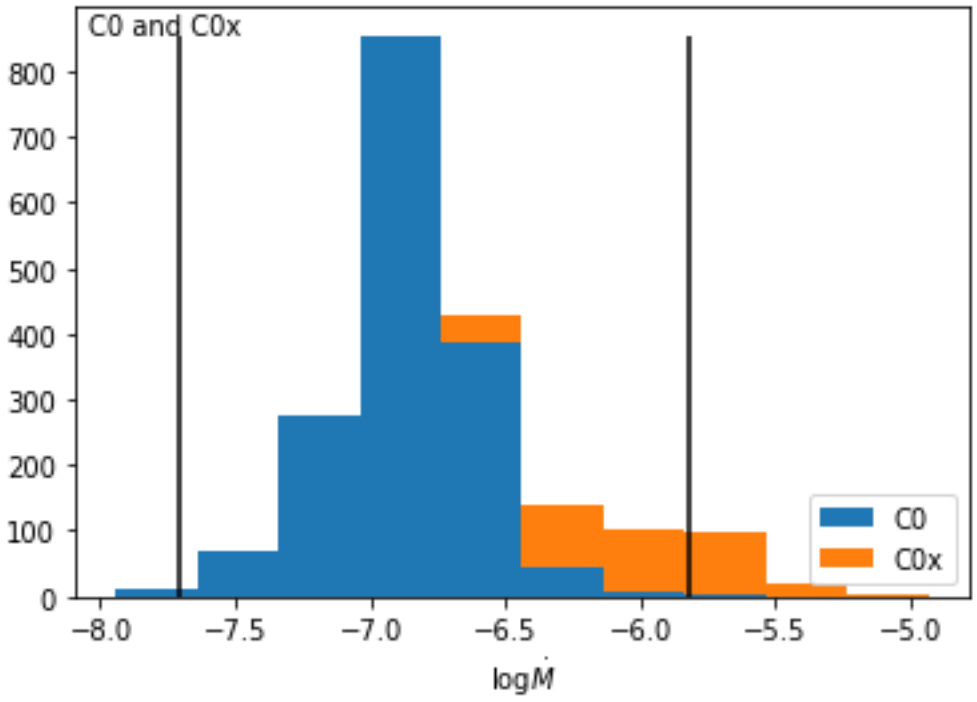

Lower bound fit:
n= 0.5
Fit quality: Slope= 2.4959025952959917
Fit significance: R^2= 0.40738409029419953
Fit standard dev: sigma= 1.4424025070954543
delta logL, delta logM, delta logMdot: [0.1731330989114248, 0.07088308013837175,
1.5528419686577806]
log A, B, C: [-27.873068316656646, -8.969064716228624, 21.907089331141627]
# Stars in Bound: 708

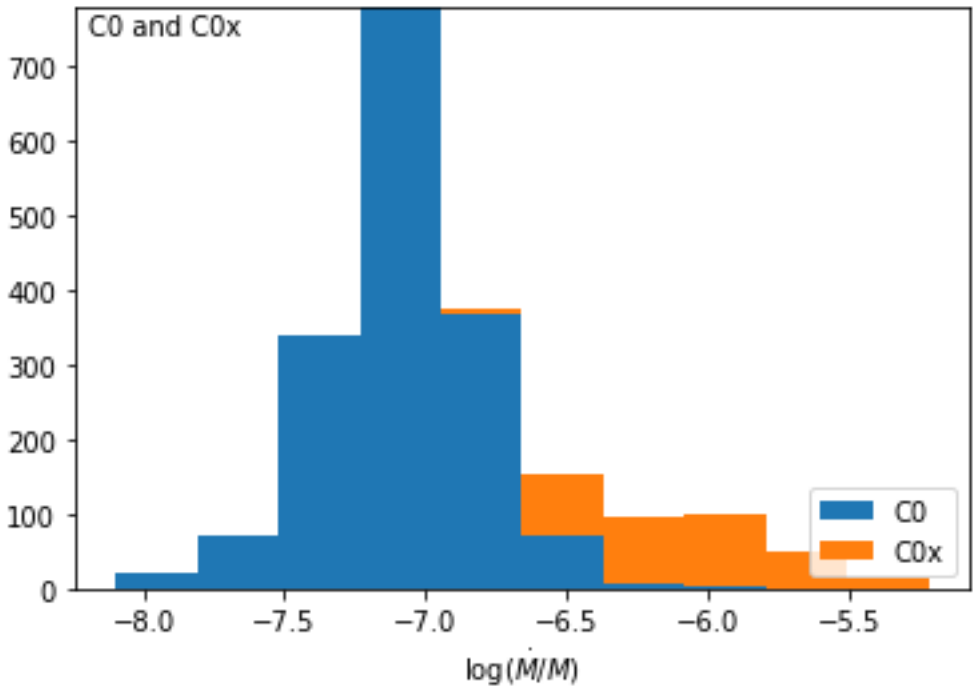

Upper bound fit:
n= 2.999900000001945
Fit quality: Slope= 0.5424388534059239
```

```
Fit significance: R^2= 0.4073840902941993
Fit standard dev: sigma= 0.3425741547903071
delta logL, delta logM, delta logMdot: [1.0387639668494377, 0.42528430421447805,
2.0248235837250323]
log A, B, C: [-11.348698132069325, -1.9492624391528568, 4.761105838281489]
# Stars in Bound: 1981
```




```
Full range: -4.933674074637962 -7.942447998945467
vlines locations: -5.819730722331125 -7.703225153259688
```





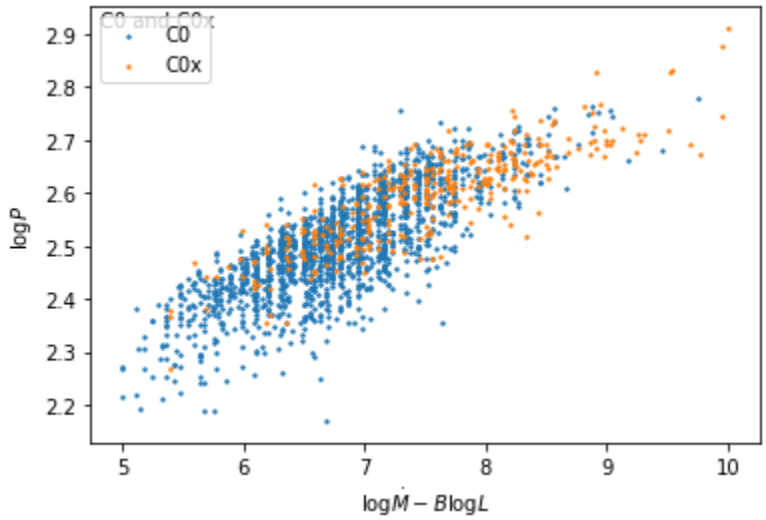
Comparison of Exponents:

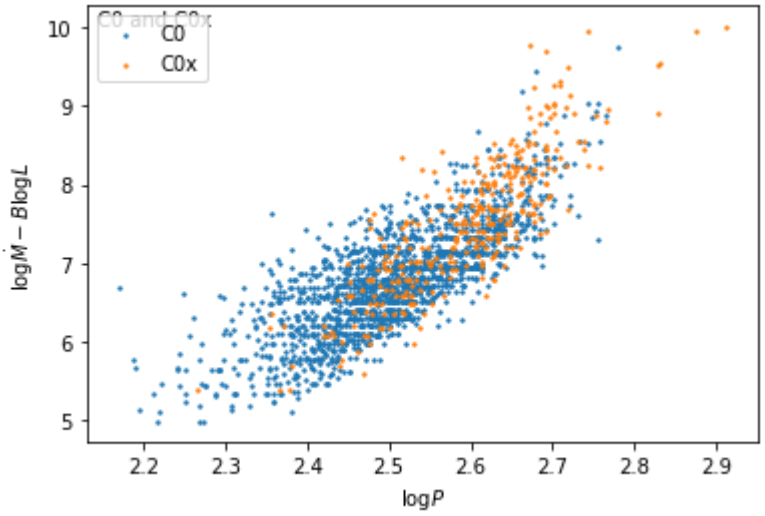
LP Strip:

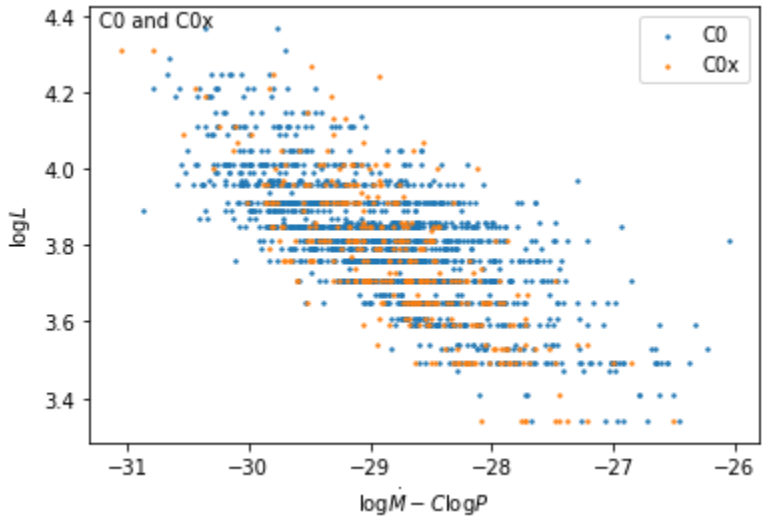
```
log A, B, C: [-41.07893702614187, 9.635192545710618, -13.910423664947327]
```

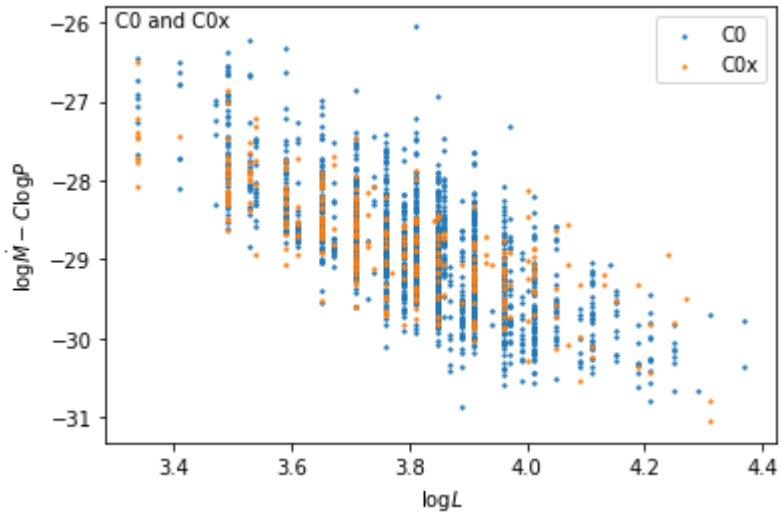
LM Strip:

```
log A, B, C: [-15.219113434240555, -3.5934733741482967, 8.777118318045444]
log A, B, C (LM Strip): [-41.06675262075109, 9.622154683267865, -
13.902542891223035]
```




C0 and C0x
C0
C0x
10
9
8
7
6
5
2.2
2.3
2.4
2.5
2.6
2.7
2.8
2.9
$\log\dot{M} - B\log L$
$\log P$


C0 and C0x
C0
C0x
4.4
4.2
4.0
3.8
3.6
3.4
−31
−30
−29
−28
−27
−26
$\log L$
$\log\dot{M} - C\log P$


C0 and C0x
C0
C0x
−26
−27
−28
−29
−30
−31
3.4
3.6
3.8
4.0
4.2
4.4
$\log\dot{M} - C\log P$
$\log L$

C1:

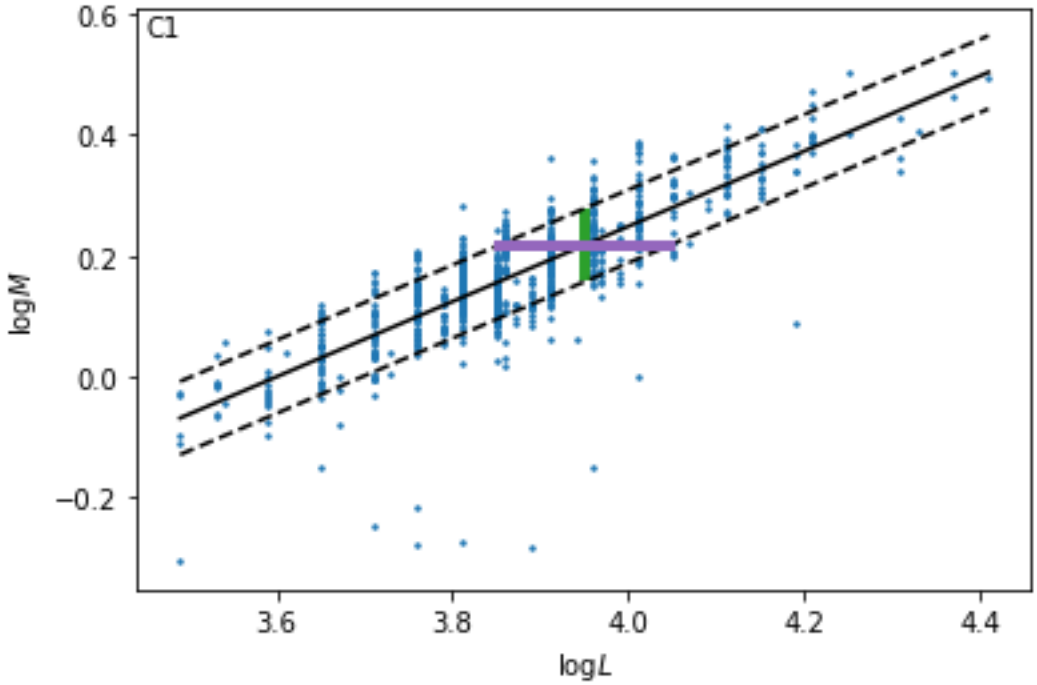
LM Strip:

deathline_fit: (array([-2.2461344 , 0.62357106]),
0.6712522002248873, Intercept   0.060517

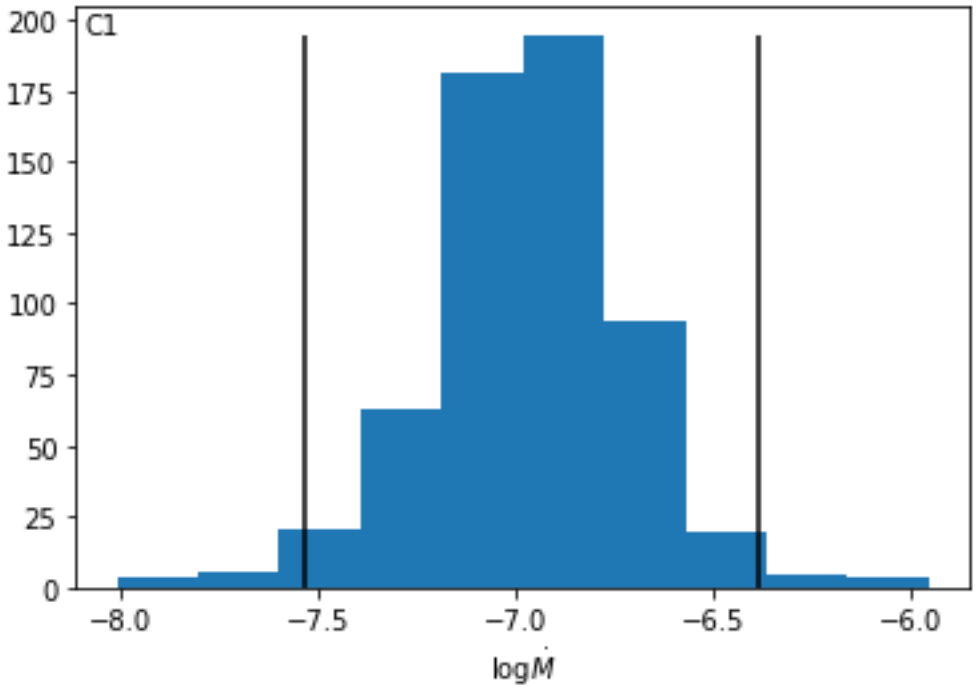
```
x           0.015635
dtype: float64,            Intercept         x
Intercept   0.944136 -0.243762
x          -0.243762  0.063021)
Optimal fit found at  6504
n= 1.1503999999999284
Fit quality: Slope= 1.0000181660375453
Fit significance: R^2= 0.24765173220056858
Fit standard dev: sigma= 0.509201851017138
delta logL, delta logM, delta logMdot: [0.19474764044216197, 0.12143899263765517,
1.1461280356782382]
log A, B, C: [-28.14202038380528, 5.88519600584648, -9.437891494192556]
# Stars in Bound: 587

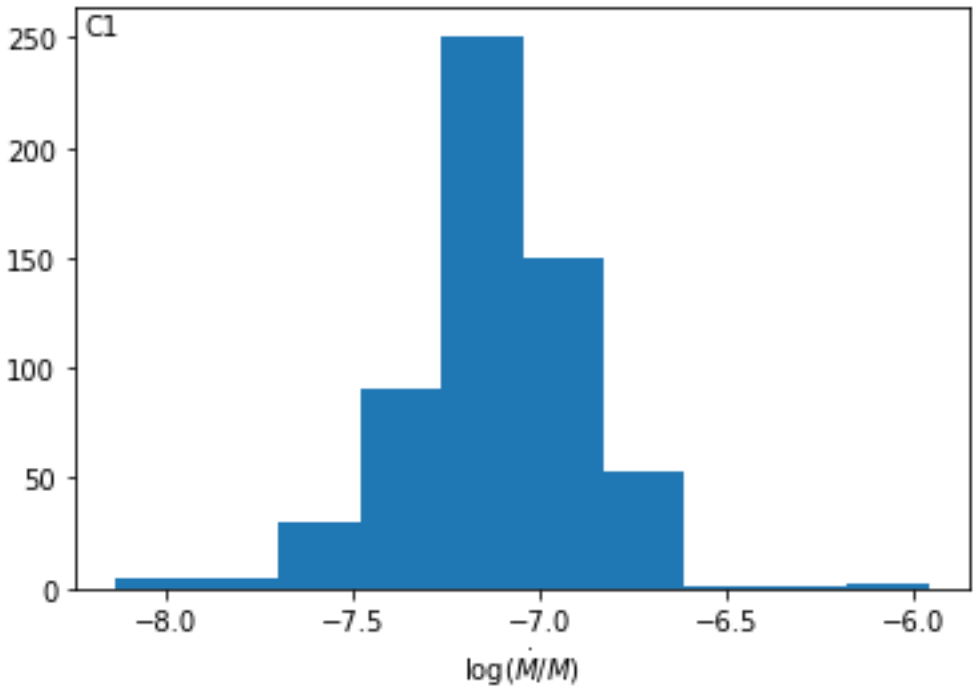

Lower bound fit:
n= 0.5
Fit quality: Slope= 2.5762157564270125
Fit significance: R^2= 0.24765173220056436
Fit standard dev: sigma= 1.3902636729215105
delta logL, delta logM, delta logMdot: [0.08464344595018058, 0.05278120333695352,
1.2833012287035501]
log A, B, C: [-61.55486794571157, 15.16125925991807, -24.31360309296088]
# Stars in Bound: 291


Upper bound fit:
n= 2.999900000001945
Fit quality: Slope= 0.4854249505701496
Fit significance: R^2= 0.24765173220056647
Fit standard dev: sigma= 0.28929934301635696
delta logL, delta logM, delta logMdot: [0.5078437470122208, 0.31667666378125925,
1.4507923159973135]
log A, B, C: [-17.23347391118014, 2.856769084059238, -4.581304787900101]
# Stars in Bound: 770
```




```
Full range: -5.955950469349115 -8.009394788576081
vlines locations: -6.389606030287687 -7.535734065965925
```





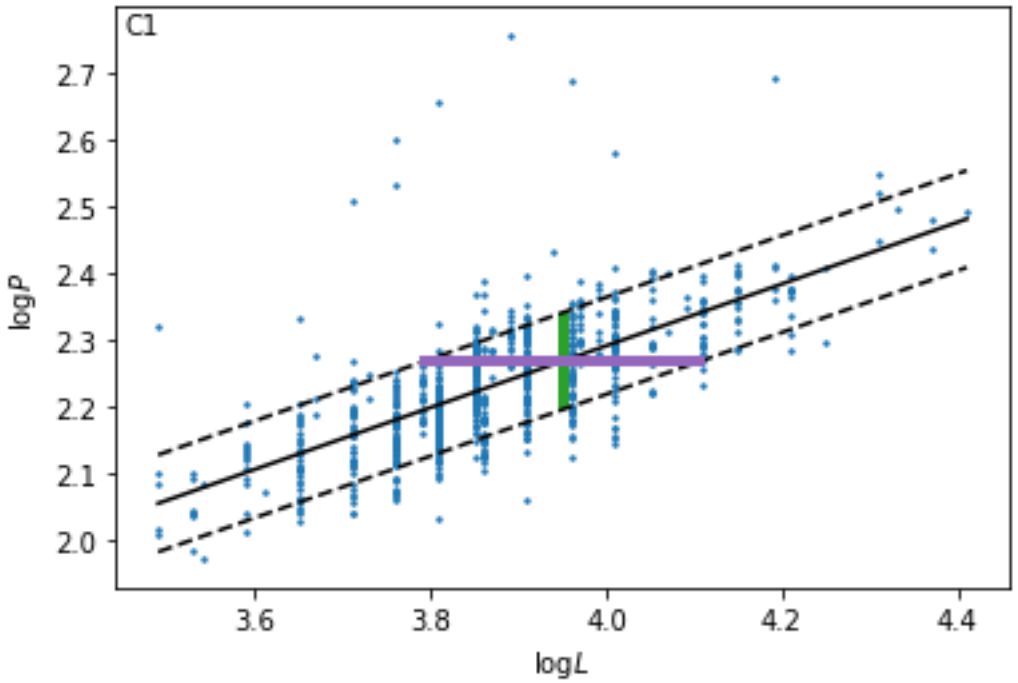
LP Strip:

deathline_fit: (array([0.43570489, 0.46380993]), 0.477803904313582, Intercept 0.067241
x 0.017373
dtype: float64, Intercept x
Intercept 0.944136 -0.243762
x -0.243762 0.063021)
Optimal fit found at 6600
n= 1.1599999999999273
Fit quality: Slope= 1.0000038344122026
Fit significance: R^2= 0.24765173220056702
Fit standard dev: sigma= 0.5091945534331314
delta logL, delta logM, delta logMdot: [0.31361833035091724, 0.1454592967944075, 1.2355284469075496]
log A, B, C: [-10.64411651430923, -3.9395925790596444, 8.49398061269228]
log A, B, C (LM Strip): [-28.151144894935417, 5.888707448080465, -9.440804631589089]
Stars in Bound: 619

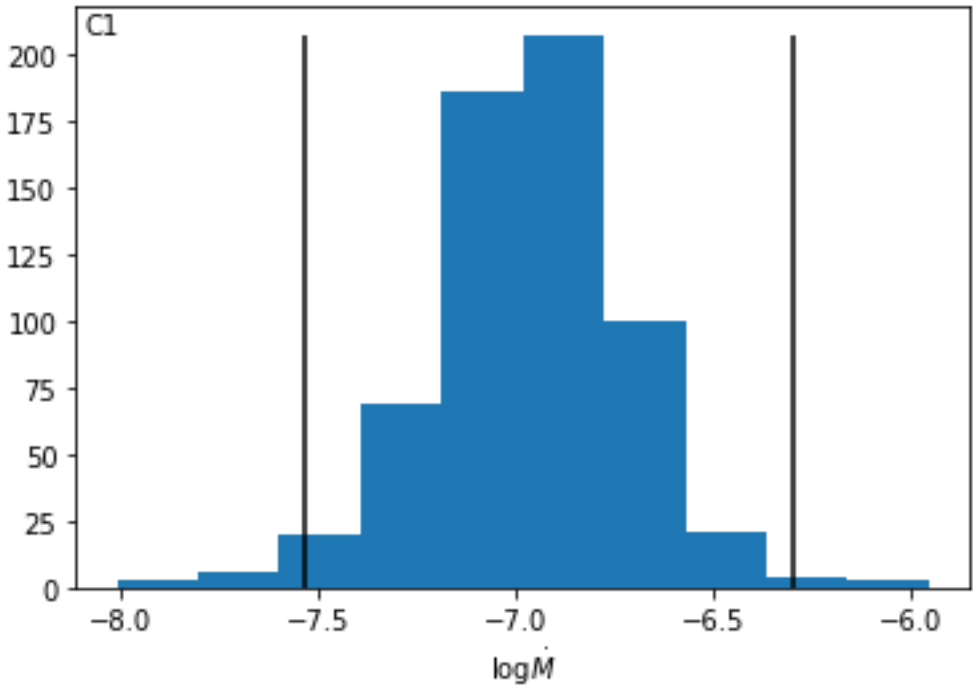
Lower bound fit:
n= 0.5
Fit quality: Slope= 2.409714057237305
Fit significance: R^2= 0.24765173220056547
Fit standard dev: sigma= 1.2942793134621795
delta logL, delta logM, delta logMdot: [0.13518031480643833, 0.06269797275621425, 1.2833012287035501]
log A, B, C: [-15.861249215646545, -9.49325521649421, 20.467986001610505]
Stars in Bound: 316

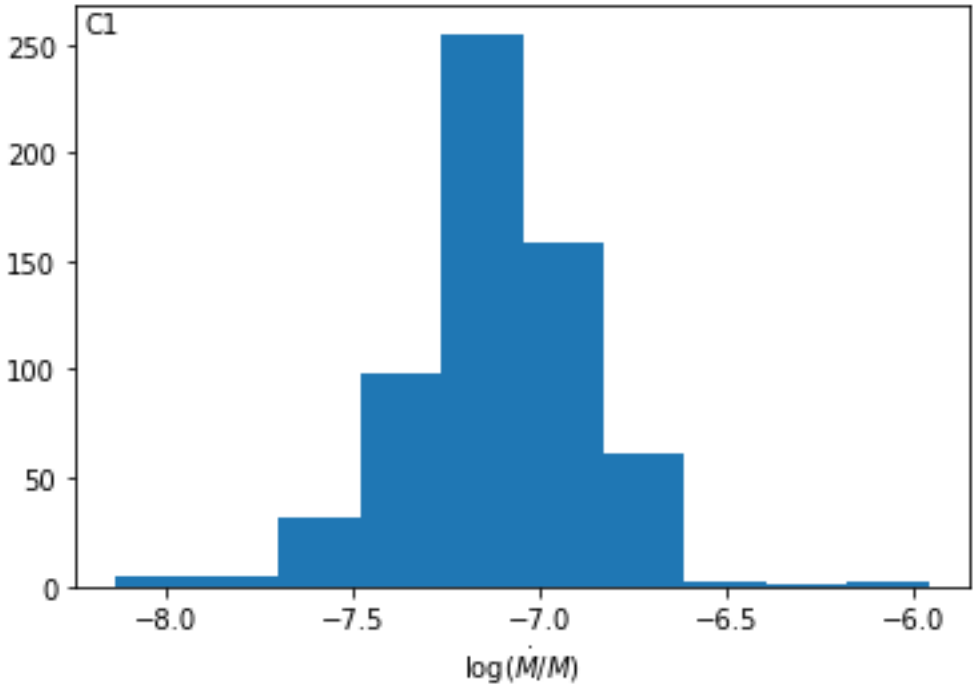
Upper bound fit:
n= 2.999900000001945
Fit quality: Slope= 0.4540517711703385

```
Fit significance: R^2= 0.2476517322005657
Fit standard dev: sigma= 0.28087672315690637
delta logL, delta logM, delta logMdot: [0.811054852776194, 0.3761752969429777,
1.4507923159973135]
log A, B, C: [-8.623627256031961, -1.7887721293218763, 3.8566921533319904]
# Stars in Bound: 771
```




```
Full range: -5.955950469349115 -8.009394788576081
vlines locations: -6.300205619058375 -7.535734065965925
```





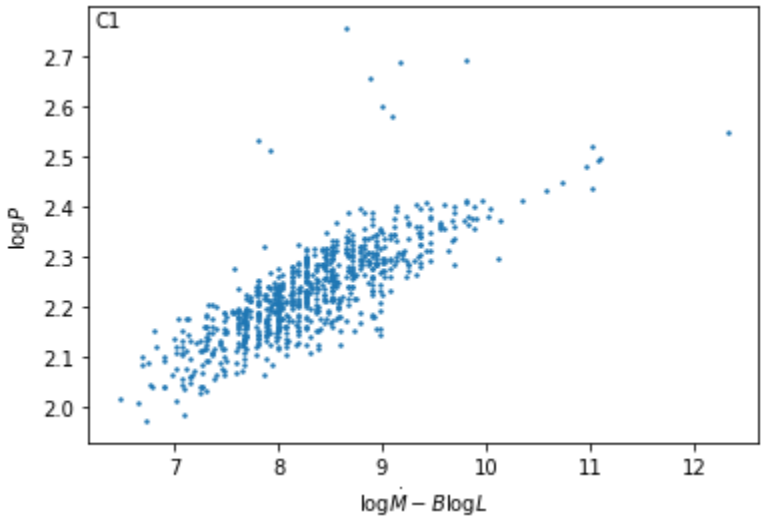
Comparison of Exponents:

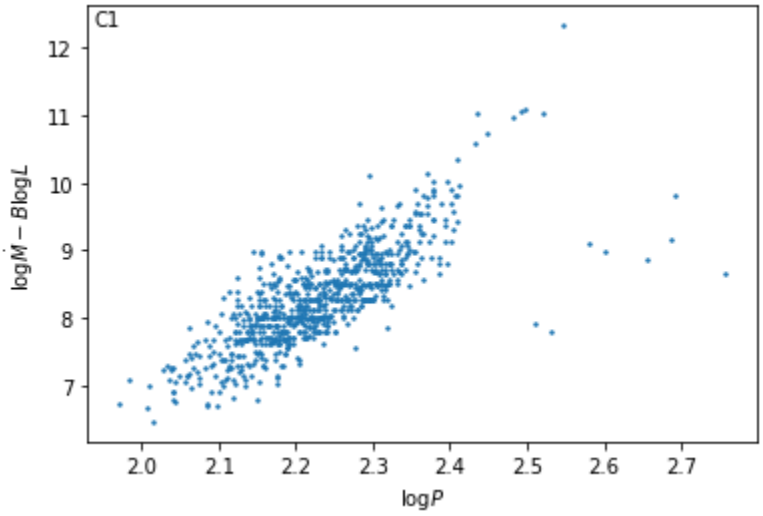
LP Strip:

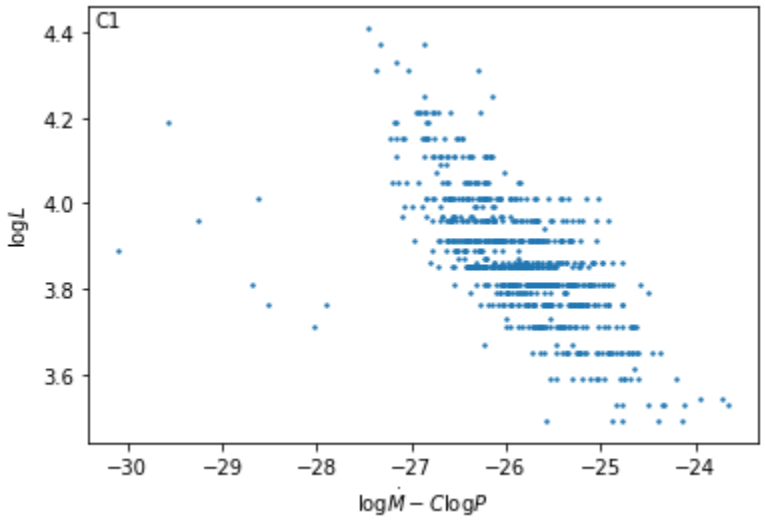
```
log A, B, C: [-28.14202038380528, 5.88519600584648, -9.437891494192556]
```

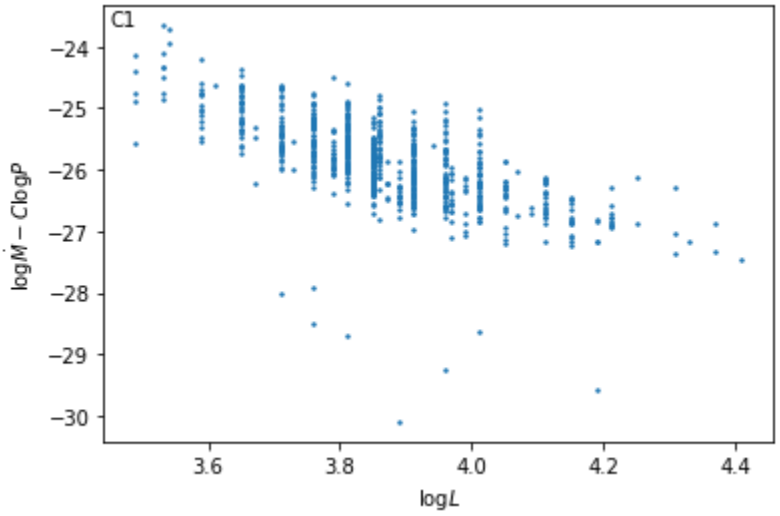
LM Strip:

```
log A, B, C: [-10.64411651430923, -3.9395925790596444, 8.49398061269228]
log A, B, C (LM Strip): [-28.151144894935417, 5.888707448080465, -
9.440804631589089]
```




C1
12
11
10
9
8
7
2.0
2.1
2.2
2.3
2.4
2.5
2.6
2.7
$\log\dot{M} - B\log L$
$\log P$


C1
4.4
4.2
4.0
3.8
3.6
−30
−29
−28
−27
−26
−25
−24
$\log L$
$\log\dot{M} - C\log P$


C1
−24
−25
−26
−27
−28
−29
−30
3.6
3.8
4.0
4.2
4.4
$\log\dot{M} - C\log P$
$\log L$